

EC-333 Microprocessor and Interfacing Techniques

Experiment # 3

Memory Read and Write operations.

1. Objective/s:

- Introduction.
- Memory Read Operation.
- Memory Write Operation.
- Flags.
- Example Codes.
- Lab Task.

Introduction:

The MOV instruction **moves data bytes between the two specified operands**. The byte specified by the second operand is copied to the location specified by the first operand. The source data byte is not affected.

A memory unit stores binary information in groups of bits called words. Data input lines provide the information to be stored into the memory, Data output lines carry the information out from the memory. The control lines Read and write specifies the direction of transfer of data. Basically, in the memory organization, there are 2^l memory locations indexing from 0 to 2^l-1 where l is the address buses. We can describe the memory in terms of the bytes using the following formula:

$$N = 2^l \text{ Bytes}$$

Where,

l is the total address buses

N is the memory in bytes

For example, some storage can be described below in terms of bytes using the above formula:

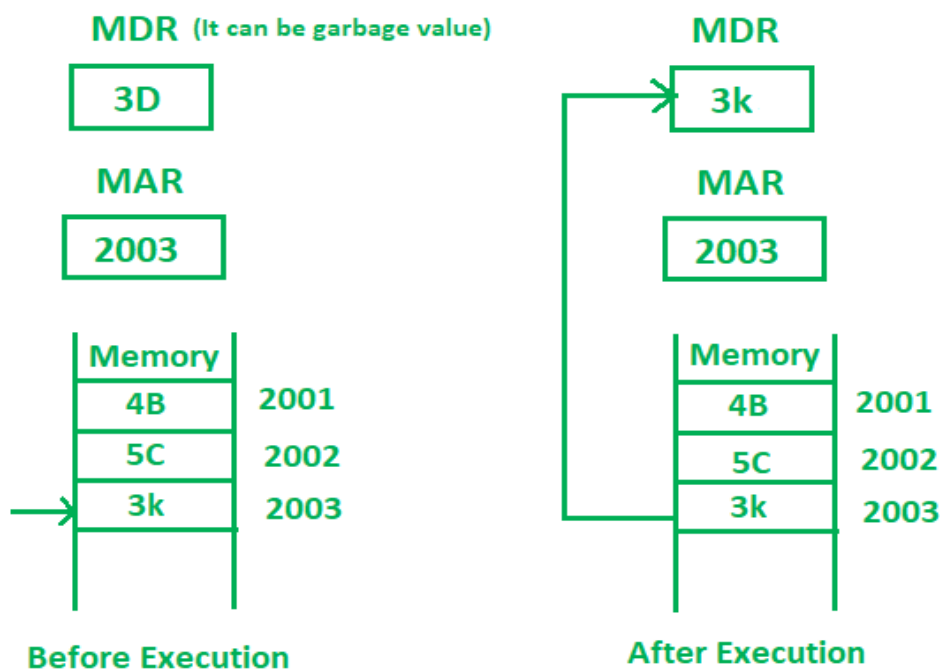
$$\begin{aligned} 1\text{ kB} &= 2^{10} \text{ Bytes} \\ 64 \text{ kB} &= 2^6 \times 2^{10} \text{ Bytes} \\ &= 2^{16} \text{ Bytes} \\ 4 \text{ GB} &= 2^2 \times 2^{10}(\text{kB}) \times 2^{10}(\text{MB}) \times 2^{10}(\text{GB}) \\ &= 2^{32} \text{ Bytes} \end{aligned}$$

Memory Address Register (MAR) is the address register which is used to store the address of the memory location where the operation is being performed.

Memory Data Register (MDR) is the data register which is used to store the data on which the operation is being performed.

Memory Read Operation:

Memory read operation transfers the desired word to address lines and activates the read control line. Description of memory read operation is given below:

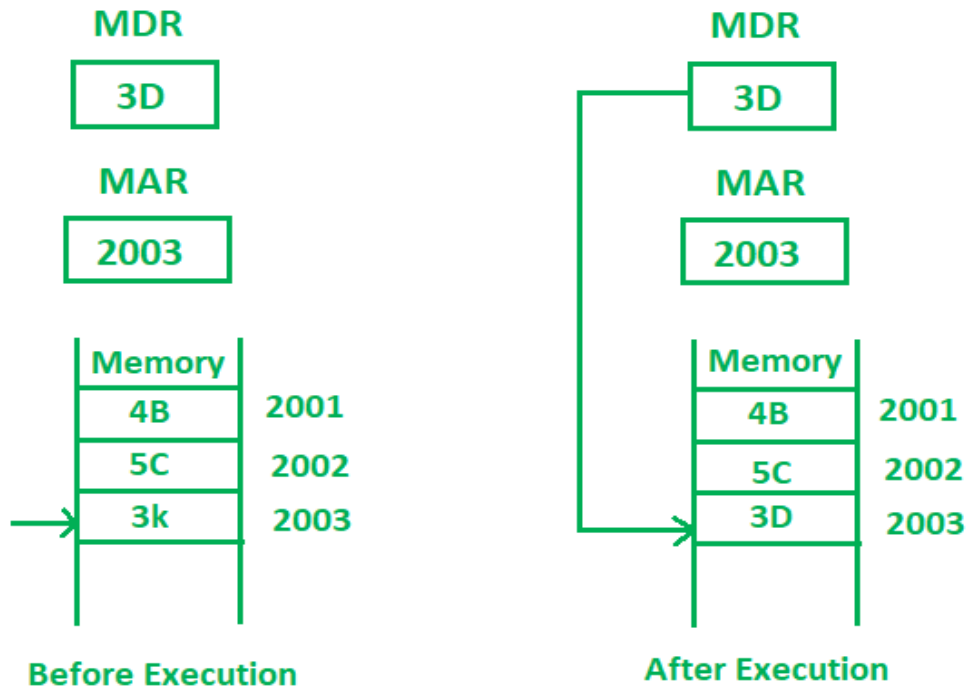


Memory Read Operation

In the above diagram initially, MDR can contain any garbage value and MAR is containing 2003 memory address. After the execution of read instruction, the data of memory location 2003 will be read and the MDR will get updated by the value of the 2003 memory location (3D).

Memory Write Operation:

Memory write operation transfers the address of the desired word to the address lines, transfers the data bits to be stored in memory to the data input lines. Then it activates the write control line. Description of the write operation is given below:



Memory Write Operation

In the above diagram, the MAR contains 2003 and MDR contains 3D. After the execution of write instruction 3D will be written at 2003 memory location.

Flags:

The flags are a set of variables in the CPU which indicate the status of various calculations and components of the CPU. Flags are used, among others, in the following contexts:

- to indicate errors
- to indicate the sign of the last calculation
- to enable a carry during arithmetic operations
- for debugging

A number of instructions perform certain tasks based on the current state of the flags. The most commonly used flags are:

CF	carry flag
ZF	zero flag
SF	sign flag
OF	overflow flag
IF	interrupt enable flag
DF	direction flag

Example:

<u>Instruction</u>	<u>Meaning</u>
MOV AX, BX	Move 16 bits from register BX to register AX
MOV AL, [1234H]	Load register AL with the byte content of the data segment memory address 1234H
MOV BL, 48	Copy 48 decimal (8-bit) (30H) into BL
MOV EAX, 100Y	Copy a 100 binary (32-bit) into EAX
MOV AH, 'A'	Copy an ASCII 'A' (8-bit) into AH
MOV AX, 'AB'	Copy an ASCII 'BA' (16-bit) into AX

Lab Task:

1. Run the code and analyze the output.

```

org 100h

Mov Ax, 0800H;
Mov DS, Ax;
Mov CL, 'A';
Mov CH, 1101_1111b;
Mov BX, 15EH;
Mov [BX], CX;
Mov AX, [BX];
ret

```

- 2. 20 bit address =081E0 and DS=0810**
- 3. Move 01FH into AI register and then add AI register value with 01H. And again add AI register value with 0E0H and check the flag status.**
- 4. Add the two numbers 100 or 50 and analyze the flags.**